

AI in road safety

Behind Luna Systems is a belief that if they can outfit bikes with sensors and intelligent software, the effect could be as transformative as installing seat belts and airbags in cars decades ago. And if the concept of ARAS for motorcycles sounds new, it's because it practically is. "While ADAS for cars is very commonplace with features such as Forward Collision Warning, or Lane Keep Assist, the same level of life-saving innovation for two-wheelers has been much slower," Fleury told me.

To that effect, at EICMA this year, Luna Systems showcased a "Smart Cluster" they built in partnership with JMO Tech and CODICO, featuring a 10.25-inch touchscreen display, along with multiple cameras: "Essentially, this cluster provides an interface for our warning systems," Fleury said.

The interface might flash an alert if a pedestrian suddenly appears in your lane, or if you're trailing an SUV too closely. The system even includes something called "AI Reverse Blind Spot Detection," which Fleury believes could be a game-changer: "It ensures that riders are aware when they are within the blind spot of a larger vehicle, such as a truck."

To meet the performance needs required to run real-time AI on a two-wheeler display, one might think you'd need an expensive, power-hungry chipset. But Luna's synergy with Qualcomm is all about making sure that's not the case. "Our partnership with Qualcomm enables us to democratise access to ARAS with its scalable solutions," Fleury said. "One of the core USPs of our team at Luna Systems is AI model optimisation and with that the ability to deliver extremely high functionality on entry-level hardware."

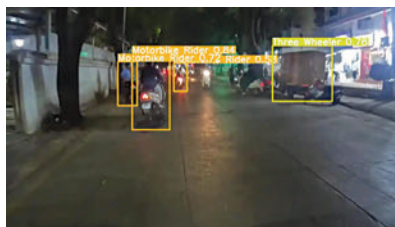
In other words, you won't need a supercomputer on your handlebars. By optimising how data is processed, they can run advanced vision algorithms on mainstream SoCs, which is crucial for cost-sensitive markets. "From a technological perspective we are also now at the point where advancements in sen-

sors, AI, and connectivity have overcome historical barriers to ARAS development for two-wheelers," he added.

This technology isn't some futuristic notion reserved for high-end sports bikes. Luna's explicit aim is to push their ARAS out at scale so everyday riders – even on lower-cost commuter bikes – can benefit. "Safety is not just a premium feature," Fleury insisted.

Making Indian two-wheeler riders more responsible

Perhaps the most intriguing and amazing aspect of Luna's solution is that it's not just about real-time collision warnings. It also provides post-ride safety analysis via an app. "Really our approach to making an impact on safety is both in-ride, as it happens risk prevention, but also giving brands the tools they need to effectively communicate/incentivise riders to be more mindful of their riding style," Fleury explained.



With the camera-based approach, the system can identify potholes, read road signs, track passing vehicles, and more, bridging a gap that radar-only systems for bikes haven't fully addressed. Radar is great at measuring distances, but it doesn't interpret the environment in detail. A well-trained camera, by contrast, can "see" and understand the difference between a child crossing the road and a stray dog meandering about. The more context, the better the warnings.

"Our solution focuses on fostering safer riding practices in-ride, such as maintaining safer distances, being aware of road positioning versus larger vehicles, being aware of the presence of pedestrians, but also helping them to perfect their riding skills overtime with AI ride coaching," Fleury reiterated. If a rider frequently brakes late, speeds

recklessly, or hovers in a truck's blind spot, the system logs that data. Then, once the ride is over, the accompanying app's AI can deliver insights – like a personal driving instructor reviewing your performance. "We also believe a ride safety coach is something which will be welcomed by parents who buy their children their first bikes," Fleury mused. Although some might find it intrusive, it's easy to see how such a tool could be a life-saver – literally.

Mass adoption in a cost-sensitive Indian market

To truly make a dent, especially in a budget-conscious Indian automotive market, you can't just equip premium bikes priced at half a million rupees – a fact that isn't lost on Fleury. "Many ARAS solutions built to date come from companies who initially developed ADAS for four-wheel vehicles. Here compute is everything and therefore cost is high. Our ambition is to scale affordable safety, to try to make the biggest impact we can on these fatalities," Fleury said.

He's right, as far as I'm concerned. Without affordability, ARAS becomes a fancy perk for well-off hobbyists, not a mainstream safety revolution that two-wheelers so desperately need. And if we're serious about that horrifying 76,000 fatalities number, we need mainstream adoption. So Luna's forging deals with manufacturers who serve the mass market, enabling them to put a relatively inexpensive cluster or an add-on module into new two-wheeler models.

"Feedback from the market is that scalability and high functionality is key," Fleury shared, referencing his meetings at EICMA with various Indian brands. Some have tried radars, others have dabbled in lower-end dashcam solutions, but Luna's pitch is that you need a robust AI approach that can handle Indian roads. **d**

